thermal, LiDAR, and inertial sensors to generate a dynamic map of its environment and make instantaneous decisions such as rerouting during adverse weather or selecting alternative flight paths when obstacles appear.

As autonomous systems continue to develop, drones are gradually becoming self-reliant aerial agents capable of coordinating in groups, collaborating with ground systems, and fulfilling mission objectives with minimal oversight. This advancement represents a significant step toward intelligent, scalable, and adaptive UAV networks, forming the foundation for future applications in logistics, urban air mobility, defense, healthcare, and smart cities.

9.2.2 AI advancements in drone navigation, object detection, and mission planning

Artificial Intelligence has significantly enhanced the core operational capabilities of drones by improving their ability to navigate environments, interpret sensory information, and plan missions intelligently. Modern UAVs employ AI-driven navigation systems such as simultaneous localization and mapping (SLAM), which allow drones to build and update maps of unfamil iar environments while determining their position without depending solely on GPS. This is especially critical in indoor settings, dense urban areas, forests, and GPS-denied environments, where traditional navigation methods are unreliable.

AI-powered object detection and recognition systems further enable drones to identify terrain features, vehicles, structures, animals, and human activity with high precision. Through computer vision and deep learning models, drones can differentiate between obstacles and safe flight paths, detect structural damage, locate disaster survivors, assess crop health, or track moving targets all in real time. These perception capabilities have transformed drones into autonomous sensing platforms with applications in